



UG Program in Electronics and Communication (Advanced Communication Technology)

Experiment No. 7 Generation (and detection) of Binary PSK

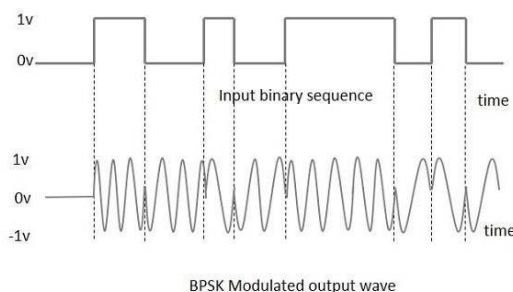
7.1 Aim: To study the Binary Phase Shift Keying Generation (and detection) technique and plot the modulated and demodulated waveforms.

7.2 Course Outcome: Compare the performances of different digital modulation techniques

7.3 Learning Objectives: Students will be able

- To understand the principles of Binary Phase Shift Keying (BPSK) modulation and demodulation techniques.
- To generate and detect BPSK signals and plot the modulated and demodulated waveforms.

7.4 Related Theory:



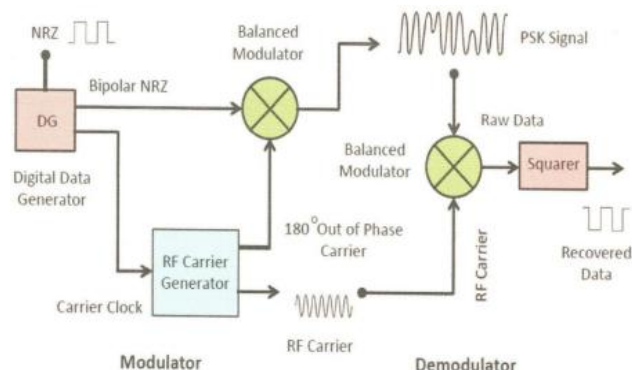
BPSK is also called 2-phase PSK or Phase Reversal Keying. In this technique, the sine wave carrier takes two phase reversals such as 0° and 180° . It is basically a Double Side Band Suppressed Carrier (DSBSC) modulation scheme, for message being the digital information.

Binary 1 and binary 0 are represented by different carrier phases each 180 degree apart. The simplest BPSK scheme uses two phases to represent the two binary digits and is known as binary phase-shift keying. The resulting transmitted signal for one bit time is

$$\begin{aligned} S(t) &= A \cos \cos (2\pi f_c t) \quad \text{for binary 1} \\ S(t) &= A \cos \cos (2\pi f_c t + \pi) \quad \text{for binary 0} \end{aligned}$$

This is regarded as the most robust digital modulation technique and is used for long distance wireless communication. It is less immune to the interference.

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BPSK Modulator Demodulator: The PSK Modulation and Demodulation System consist of following sections.

1. Digital Data Generator
2. RF Carrier generator
3. PSK modulator.
4. PSK demodulator.
5. Data squarer section
6. Power supply

Digital Data Generator section: To generate PSK signal input, digital modulating signal is required. This section generates this digital data signal. It consists of many digital ICs. The carrier clock of 1.28 MHz is generated by IC 74HC04. This carrier clock frequency is divided by 16 to produce bit clock (Bk) of 80 KHz. This Bk frequency is divided by 8 to generate Word clock (Wk) of 10KHz. Then a word of 8-bit data is generated using D-FFs and eight data generating push switches. By pushing different push switches different data word can be generated. This data word is called as roc data. Then this NRZ data is converted to produce bipolar NRZ data by IC 4052. Each push switch controls one bit from LSB to MSB viz. D1 to D8. The Bipolar NRZ data is used is then given to PSK modulator.

RF Carrier Generator: The function of this section is to generate phase synchronized carrier signal with respect to modulating digital NRZ input. It converts 1.28 MHz TTL clock signal Ck into 1.28MHz sine wave carrier signal using pie filter made of 100uH coil and two 100 pf capacitor. Then 180° out of phase sine wave signal referred as el is generated by transformer circuit. This signal is given as carrier signal to balanced modulator as one input. The phase control preset pot controls the degree of phase shift of this signal from carrier



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phase. The gain control controls the amplitudes of carrier signal. PSK modulator using balanced modulators: IC 1496 is used as balanced modulator. The modulating Bipolar NRZ signal is connected at pin 8. The carrier signal el is applied at pin 1 through buffer transistor BC548. The modulated outputs available at pin 12 and 6 are then balanced amplified by four transistors (DC558 and BC548). The final balanced modulated output is available at PSK O/P terminal. Presets 10K and 1K at pin 14 and 1 are adjusted balancing input signal and carrier signal. Preset 1K at output is used to adjust output zero DC level. The preset 2K at output decides output level of product detector.

PSK Demodulator using balanced modulators: The balanced modulator is used to demodulate PSK signal. The balanced modulated circuit is same as above. Two input signals given are PSK modulated signal from modulator section. The output of balanced modulator is raw data signal for NRZ signal. This raw data signal is then given to data squarer section.

Data Recovery and squarer section: IC 311 is used to convert raw data into pure transient free pure digital data. The threshold Zero level is set by 10K preset. The output is available at pin 7. The recovered NRZ signal is available at output terminals.

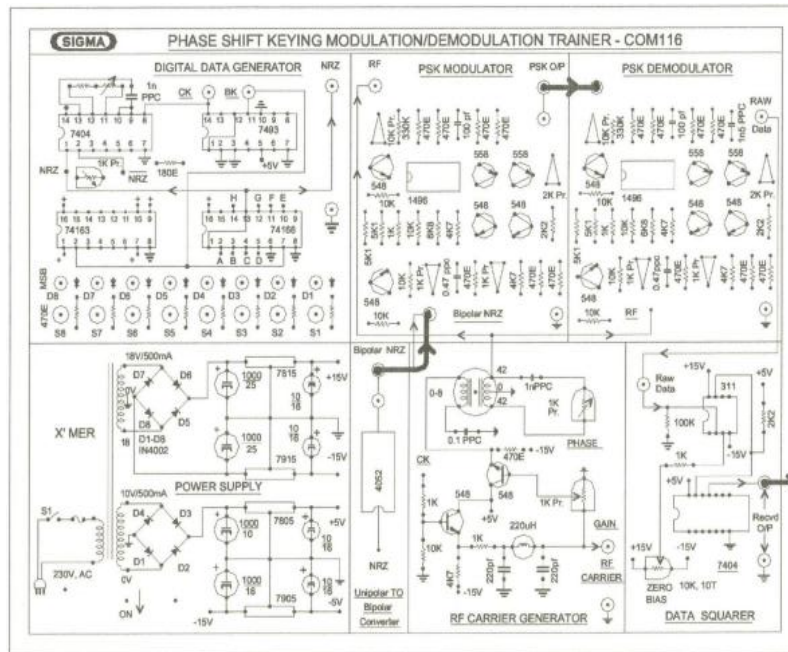
Power supply section: The regulated power supply is used for different supply voltages. Following output D.C. Voltages are required to operate FSK Modulation demodulation system:

- +15V 250mA,
- 15V 250mA,
- + 5V 250mA.

Three terminal regulators are used for different output voltages i.e. IC 7805 for +5V, IC 7815 for +15V, and IC 7415 for -15V. These ICs are supplied different dc input voltages by two Bridge rectifiers consisting of DI-D4 and D5-D8 & two 1000/25 EC and 1000/10 EC. The capacitors at each input & each output are for filtering purpose

Circuit Details:

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BPSK Circuit Connection Diagram

Procedure:

1. First apply any digital word by pushing toggle switches in data generator section e.g., 11110000.
2. This digital word is NRZ data, which can be observed at NRZ DATA output terminal.
3. Connect CRO Channel-1 at NRZ DATA (NRZ) socket and observe it.
4. Connect CRO Channel-1 at RF Carrier socket (in Carrier generator section) and observe it.
5. Connect CRO Channel-1 at Bipolar NRZ socket and observe it.
6. Connect CRO Channel 1 at RF Carrier and CRO Channel 2 at $\Phi 1$ signal and observe their 180° out of phase relationship.
7. Now connect CRO Channel 2 at the PSK output of PSK modulator.

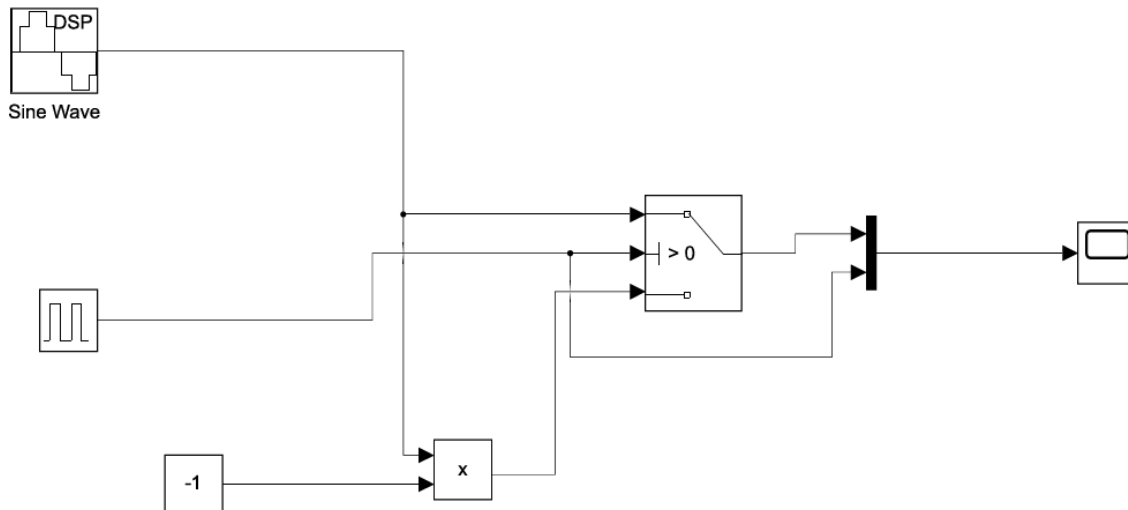


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8. Connect CRO channel 1 at Bipolar NRZ terminal. Observe PSK output.
9. Observe recovered raw data signal at output of PSK demodulator.
10. Observe received pure NRZ data at the output data squarer.
11. Apply different digital word by pushing push switches in data generator and observe recovered signal

7.5 Output:

Simulation Circuit:





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Simulation Parameters:

Block Parameters: Sine Wave

Sine Wave (mask) (link)

Output samples of a sinusoid. To generate more than one sinusoid simultaneously, enter a vector of values for the Amplitude, Frequency, and Phase offset parameters.

Main Data Types

Amplitude: 1

Frequency (Hz): 10

Phase offset (rad): 0

Sample mode: Discrete

Output complexity: Real

Computation method: Trigonometric fcn

Sample time: 1/1000 0.001

Samples per frame: 1

Resetting states when re-enabled: Restart at time zero

OK Cancel Help Apply

BLOCK

Stop Time 2.0

Block Parameters: Pulse Generator

Pulse Generator

Output pulses:

```
if (t >= PhaseDelay) && Pulse is on
    Y(t) = Amplitude
else
    Y(t) = 0
end
```

Pulse type determines the computational technique used.

Time-based is recommended for use with a variable step solver, while Sample-based is recommended for use with a fixed step solver or within a discrete portion of a model using a variable step solver.

Parameters

Pulse type: Time based

Time (t): Use simulation time

Amplitude: 1

Period (secs): 1

Pulse Width (% of period): 50

Phase delay (secs): 0

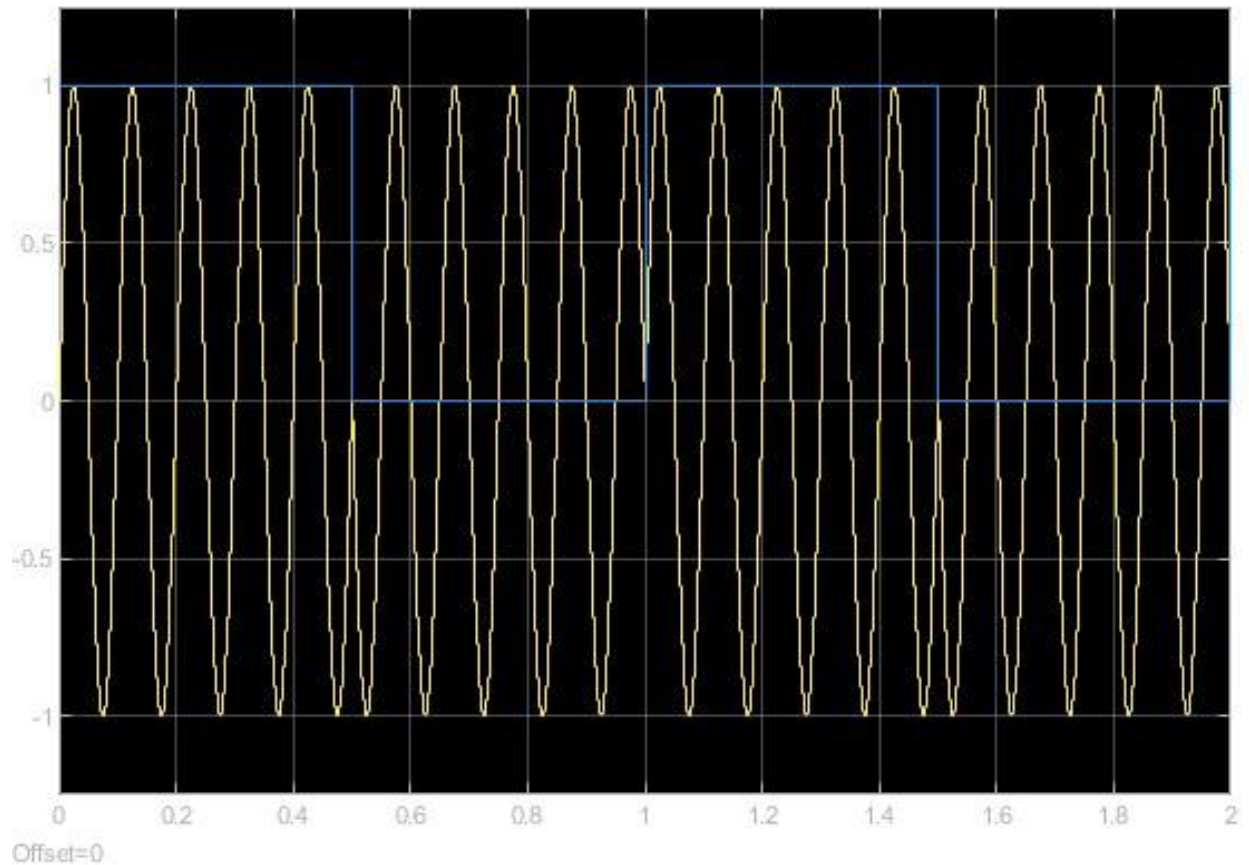
☒ Interpret vector parameters as 1-D

OK Cancel Help Apply



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Simulated Output Waveform:



7.6 Conclusion:

The experiment helped in understanding the working principle of Binary Phase Shift Keying (BPSK) modulation and demodulation. The BPSK signal was generated by changing the phase of the carrier according to the binary input data. The modulated signal was successfully demodulated to recover the original binary signal. Thus, the objectives of generating, detecting, and observing BPSK waveforms were successfully achieved.



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7.7 Questions:

1 What is Binary Phase Shift Keying (BPSK), and how does it work?

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Q1. What is Binary Phase Shift Keying (BPSK), and how does it work?

→ Binary Phase Shift Keying (BPSK) is a digital modulation technique used in digital communication system to transmit binary data.

- In BPSK, the phase of the carrier signal is changed according to the binary data.
- Two phases of the carrier wave are used to represent the binary digits:
 - At 0° phase $\rightarrow$ represents binary '0'.
 - At 180° phase $\rightarrow$ represents binary '1'.

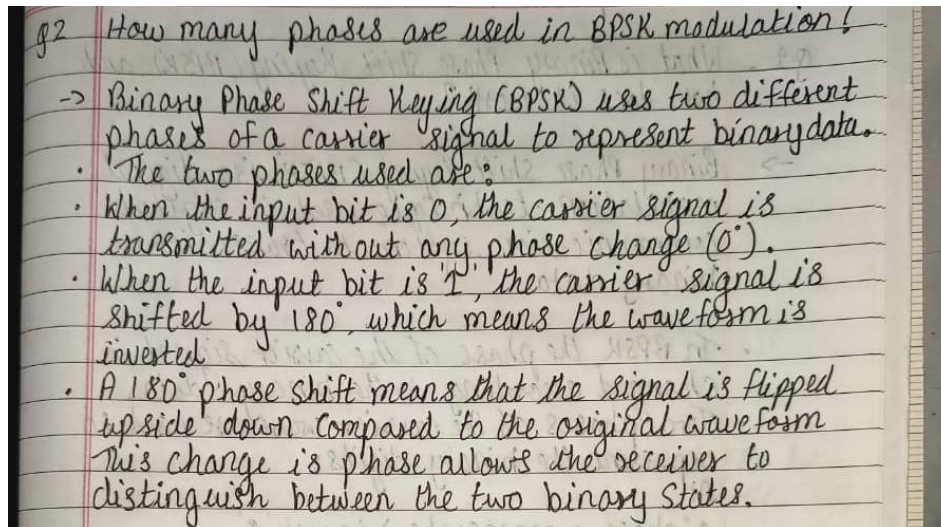
Working:

- A carrier signal is generated.
- The binary input data controls the phase of the carrier.
- When the input data bits changes for '0' to '1' the phase of the signal shift by 180° .
- At the receiver, the phase of the received signal is detected and converted back into binary data.

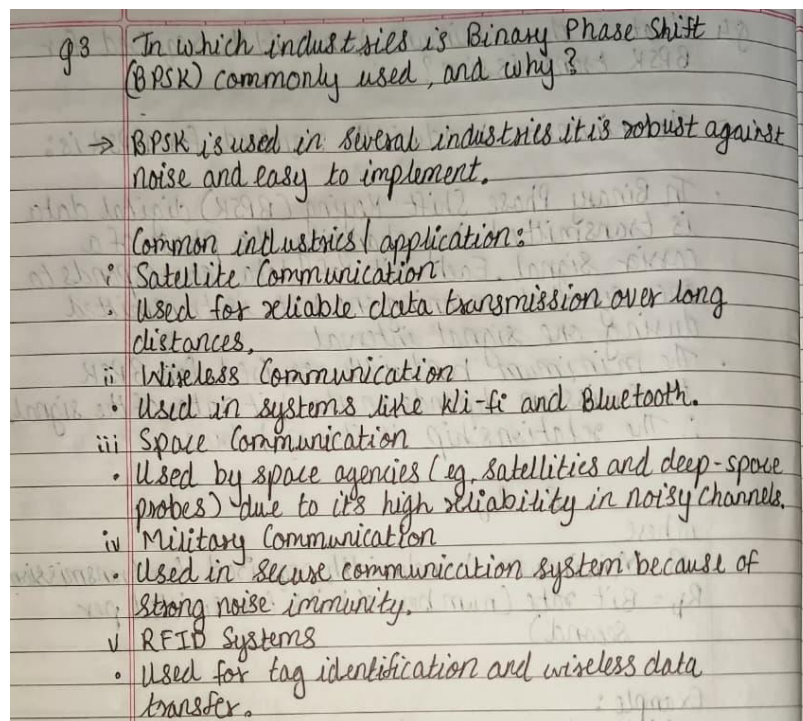
BPSK is simple, reliable and resistant to noise, making it widely used in communication system.

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2 How many phases are used in BPSK modulation?



3 In which industries is Binary Phase Shift Keying (BPSK) commonly used, and why?





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4 What is the minimum bandwidth required for BPSK transmission?

Q4 What is the minimum bandwidth required for BPSK transmission?

→ The minimum bandwidth required for BPSK is:

- In Binary Phase Shift Keying (BPSK) digital data is transmitted by changing the phase of a carrier signal. Each bit of data corresponds to one symbol, meaning one bit is transmitted during one signal interval.
- The minimum bandwidth required for BPSK transmission depends on the bit rate of the signal. ∴ The relationship is shown below:

$$B = R_b$$

where,

B = Minimum bandwidth required for transmission
 R_b = Bit rate (number of bits transmitted per second)

Example:
If bit rate = 10 Kbps, then minimum bandwidth ≈ 10 KHz.